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A Comparative Study of Tree-based Models for Healthcare Risk Prediction: Cross-validation and SHAP-based Stability Analysis

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I. Research Summary

This study compares AdaBoost, XGBoost, and LightGBM for binary healthcare risk prediction on three datasets with different sizes, feature representations, and class prevalences. A common protocol is used: stratified 80/20 development–test separation, persisted stratified five-fold partitions, fold-local preprocessing, fixed model configurations, and an untouched hold-out evaluation. Predictive performance is measured with ROC-AUC, PR-AUC, Recall, and F1-score, with PR-AUC as the primary ranking metric.

Under this protocol, XGBoost achieved the best overall cross-dataset rank (mean rank 1.0000), with mean ROC-AUC 0.8844, PR-AUC 0.5730, and Recall 0.8202. AdaBoost achieved the highest cross-dataset mean F1-score (0.5392), showing that the preferred model depends on the evaluation objective. Predictive variability was substantially higher on the smaller Heart Failure Prediction dataset: for example, XGBoost's ROC-AUC standard deviation was 0.0332, compared with 0.0031 and 0.0026 on the two larger datasets.

For explanation stability, fold-level SHAP rankings were compared using Kendall's τ, Spearman's ρ, and top-10 Jaccard similarity. Across the nine dataset–model combinations, mean Spearman correlation ranged from 0.9324 to 0.9921, while top-10 Jaccard similarity ranged from 0.7758 to 1.0000. These results indicate generally consistent explanations, although complete-ranking agreement does not always imply identical membership of the ten most important features.

Overall, the study shows that XGBoost was strongest under the selected ranking-oriented protocol, but no model dominated every metric and dataset. The contribution is a controlled comparison that evaluates predictive strength, predictive stability, and explanation stability together; the conclusions remain limited to the three datasets, fixed configurations, and evaluation protocol used here.

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II. Introduction and Research Questions

II.1. Research Context

Healthcare risk prediction has become an important application of machine learning, particularly with the increasing availability of structured healthcare data. Machine learning models can be used to identify patterns in demographic, clinical, and physiological features and estimate the risk of developing a disease or experiencing an adverse health outcome. Such predictive models may provide useful decision-support information for healthcare-related analysis.

Among machine learning approaches for structured data, tree-based models are widely used because they can capture nonlinear relationships and feature interactions while requiring relatively few assumptions about the underlying data distribution. In particular, boosting-based models such as AdaBoost, XGBoost, and LightGBM have demonstrated strong predictive capabilities and are commonly applied to classification problems.

However, healthcare datasets can differ substantially in terms of sample size, feature distributions, missing values, and class imbalance. As a result, the performance of a particular model may vary across datasets. Evaluating multiple models across multiple healthcare datasets under a consistent experimental protocol can therefore provide a more comprehensive understanding of their behavior.

II.2. Research Problem

Although tree-based models can achieve strong predictive performance, comparing models based solely on a single train-test split or a single performance metric may provide an incomplete view of their reliability. A model that performs well on one particular split may exhibit considerable variation when trained on different subsets of the data.

Cross-validation provides a systematic approach for evaluating this variability by repeatedly training and evaluating models on different data partitions. Therefore, model comparison should consider not only average predictive performance but also the consistency of performance across cross-validation folds.

Furthermore, predictive performance does not indicate why a model produces its predictions. In healthcare-related machine learning, identifying influential features can provide additional insight into model behavior. SHapley Additive exPlanations (SHAP) can be used to quantify the contribution of individual features to model predictions. However, feature importance obtained from a single training run may not necessarily remain consistent when the training data changes.

This raises a second stability concern: whether the explanations produced by a model remain consistent across different cross-validation folds. A feature that appears highly important in one fold but substantially decreases in importance in other folds may indicate that the corresponding interpretation is sensitive to the training data.

Therefore, this study considers both predictive stability and explanation stability when comparing tree-based models for healthcare risk prediction.

II.3. Research Questions

Based on the research problem described above, this study investigates the following research questions:

- **RQ1:** How do AdaBoost, XGBoost, and LightGBM compare in healthcare risk prediction across multiple datasets?
- **RQ2:** How stable is the predictive performance of these models across cross-validation folds?
- **RQ3:** How stable are SHAP-based feature importance rankings across cross-validation folds?

II.4. Contributions

This study makes the following contributions:

- **Systematic comparison of boosting-based tree models.** We provide a controlled empirical comparison of AdaBoost, XGBoost, and LightGBM for healthcare risk prediction across multiple datasets using a consistent experimental protocol.
- **Evaluation of predictive stability.** In addition to reporting predictive performance, we analyze the variability of model performance across cross-validation folds to assess the consistency of the models.
- **Analysis of SHAP explanation stability.** We investigate the stability of SHAP-based feature importance rankings across cross-validation folds, providing an empirical assessment of whether model explanations remain consistent under different training subsets.
- **Multi-dataset evaluation.** By evaluating the models across multiple healthcare datasets, the study examines whether observed performance and explanation patterns are consistent across different data characteristics.
- **Reproducible experimental framework.** The experiments are conducted using a standardized and reproducible pipeline, enabling consistent comparison across models, datasets, and cross-validation folds.

III. Related Work

III.1. Boosting-Based Tree Ensembles

Boosting constructs an ensemble sequentially so that later learners focus on errors made by earlier learners. This makes boosting well suited to structured tabular prediction because non-linear effects and feature interactions can be captured without requiring a single highly complex tree. The three models used in this study represent different generations of boosting-based tree methods: AdaBoost provides a classical boosting baseline, while XGBoost and LightGBM are modern gradient-boosted decision-tree systems designed for stronger predictive performance and computational efficiency.

III.2. AdaBoost

AdaBoost is one of the foundational boosting algorithms. It updates observation weights after each boosting round, increasing emphasis on samples that were previously misclassified, and combines the weak learners into a weighted final classifier [1]. In this study, AdaBoost serves as the classical baseline and uses decision stumps as weak learners. Keeping the base learners simple provides a clear comparison with the deeper and more heavily regularized gradient-boosting systems used by XGBoost and LightGBM.

III.3. XGBoost

XGBoost extends gradient tree boosting with both statistical regularization and systems-level optimizations for scalable learning. Its design includes a regularized objective, sparsity-aware split handling, approximate tree learning, and efficient parallel computation [2]. It also exposes controls such as tree depth, shrinkage, row subsampling, column subsampling, and regularization. These controls are useful for the present study because the three healthcare datasets differ substantially in sample size, dimensionality, and class prevalence, while the same fixed model configuration must remain reproducible across datasets.

III.4. LightGBM

LightGBM is a gradient-boosted decision-tree framework designed for high computational efficiency. Ke et al. introduced Gradient-based One-Side Sampling (GOSS) and Exclusive Feature Bundling (EFB) to reduce computational cost while preserving predictive quality on large and high-dimensional datasets [3]. LightGBM also grows trees leaf-wise, allowing it to fit complex nonlinear relationships efficiently. Because leaf-wise growth can increase model complexity, parameters such as the number of leaves, maximum depth, minimum child samples, learning rate, subsampling, and regularization are controlled explicitly in the experimental configuration.

III.5. Explainability and Stability of Tree Models

Predictive performance alone does not show which variables drive model outputs. SHAP (SHapley Additive exPlanations) provides additive feature attributions for individual predictions and establishes desirable consistency properties for feature attribution [4]. Tree-specific SHAP algorithms make these explanations computationally practical for tree ensembles, and prior work has demonstrated their use in medical prediction settings [5]. This motivates evaluating not only predictive performance across cross-validation folds but also whether feature-importance rankings remain consistent as the training sample changes.

However, healthcare prediction studies are difficult to compare directly when they use different datasets, preprocessing choices, class-balance strategies, train–test partitions, or evaluation metrics. A reported advantage for one model may therefore reflect the experimental setup rather than a stable property of the algorithm. Explainability studies also commonly report a single global ranking, which does not establish whether that ranking persists when the training sample changes. These limitations leave two related questions open: whether model performance is stable across folds and datasets, and whether the features used to explain that performance are stable under the same changes.

III.6. Research Gap and Motivation

The gap addressed by this study is therefore methodological rather than a claim that one boosting algorithm is universally best. AdaBoost, XGBoost, and LightGBM are evaluated under one documented preprocessing, fold-generation, metric, and hold-out protocol across three healthcare datasets. The same persisted folds are used for model evaluation and SHAP analysis, while preprocessing is fitted only on each fold's training rows. The study consequently separates average predictive strength (RQ1), fold-to-fold predictive variability (RQ2), and fold-to-fold explanation variability (RQ3). This controlled design does not eliminate differences between the datasets, but it makes those differences explicit rather than conflating them with different evaluation procedures.

IV. Methodology and Research Architecture

This section describes the research architecture used to answer RQ1–RQ3. The architecture separates data preparation, predictive modeling, evaluation, and explanation analysis while keeping the dataset partitions and experiment configuration consistent across all model families.

IV.1. General Research Pipeline

The general pipeline consists of the following stages:

1. **Dataset preparation.** The raw healthcare datasets are loaded, normalized, and cleaned. Dataset-specific operations include target cleaning, conversion of invalid zero values to missing values, removal of exact duplicates where appropriate, and identification of numeric and categorical features.

2. **Development–test separation.** Each dataset is divided into a stratified 80/20 development–test split using the configured random seed. The test partition is kept untouched until all cross-validation comparisons are complete.

3. **Fold construction.** The development partition is divided into five stratified folds. The fold assignment is persisted in `kfold_indices.csv` so that AdaBoost, XGBoost, LightGBM, and SHAP analysis use the same validation partitions.

4. **Fold-local preprocessing.** For every fold, outlier bounds, imputers, encoders, scalers, and feature selectors are fitted only on the fold-training rows. The fitted transformations are then applied to the corresponding validation rows.

5. **Model fitting and prediction.** A fresh model object is created for every dataset, model, and fold. Balanced sample weights are computed from the fold-training labels, and the fitted model produces positive-class probabilities for the validation rows.

6. **Predictive evaluation.** ROC-AUC, PR-AUC, Recall, and F1-score are computed for every fold. Mean, standard deviation, minimum, maximum, and range are then aggregated for RQ1 and RQ2.

7. **Hold-out evaluation.** After cross-validation, preprocessing is fitted on the complete development partition, each model is refitted, and the final model is evaluated once on the untouched test partition.

8. **SHAP analysis.** The same fold assignments are reused for fold-wise SHAP analysis. Feature rankings are compared across folds using Kendall correlation, Spearman correlation, and top-10 Jaccard similarity for RQ3.

The experiment-level settings, including the random seed, split configuration, metrics, model parameters, and SHAP parameters, are stored in `config.yaml`. The dataset registry in `src/config.py` stores dataset-specific metadata, while the experiment configuration loader validates the YAML configuration before execution.

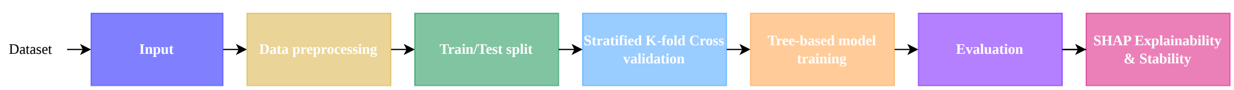

Figure 1: General research pipeline for data preparation, model evaluation, and SHAP-based stability analysis.

IV.2. Detailed Experimental Pipeline

The detailed pipeline is implemented through the preprocessing entry point, the model experiment runner, and the SHAP experiment runner. The preprocessing entry point creates reproducible development/test artifacts and the persisted fold assignment. The model runner consumes the raw development rows together with the fold manifest so that every fold can fit its own preprocessing state before training a model. The SHAP runner follows the same fold sequence and explains validation observations using a background derived from the fold-training data.

The detailed implementation pipeline is shown in the following figure.

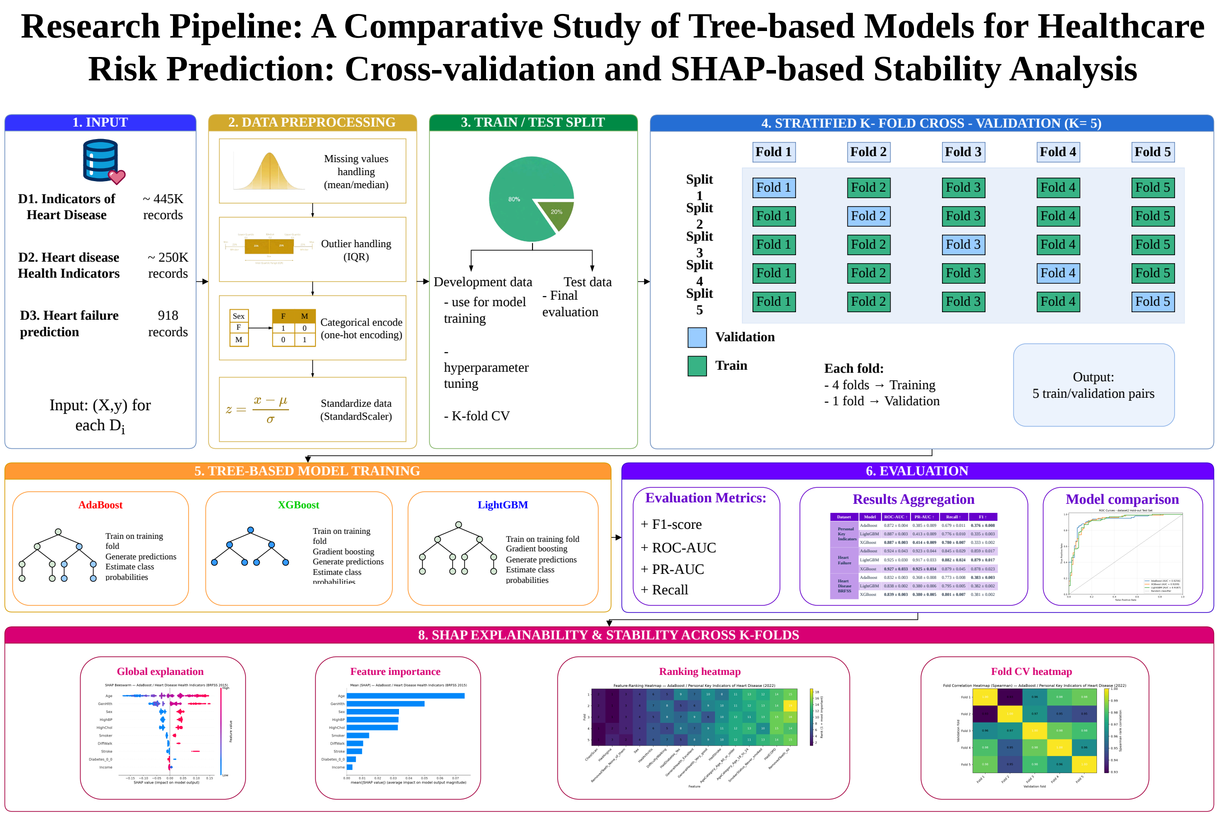

Figure 2: Detailed implementation pipeline showing fold-local preprocessing, model fitting, hold-out evaluation, and SHAP stability analysis.

IV.3. Research Architecture and Reproducibility

The architecture deliberately uses fixed model configurations instead of tuning separate configurations for each dataset. This keeps the primary comparison focused on model behavior and fold variability rather than on the variability introduced by repeated hyperparameter searches. It also means that nested cross-validation is not required for the current study design: no model or hyperparameter is selected from the same folds used to report the final stability estimates.

All generated outputs are organized by analysis stage. The model runner writes fold-level metrics, cross-validation summaries, hold-out metrics, rankings, and experiment metadata. The SHAP runner writes fold-level importance values, pairwise stability metrics, consensus rankings, and explainability plots. This separation makes the provenance of each result explicit and allows the report tables and figures to be regenerated from the configured experiment.

V. Data and Experimental Design

V.1. Dataset Description

Three publicly available Kaggle datasets are used, all framed around the same binary target: whether a respondent or patient has (or has had) heart disease or a heart attack. The datasets differ substantially in size, feature type, and data-quality characteristics, which is intentional: comparing AdaBoost, XGBoost, and LightGBM across three heterogeneous datasets, rather than a single one, lets RQ1–RQ3 be answered under different sample sizes, feature compositions, and class-imbalance regimes rather than for one dataset in isolation. Each dataset is loaded directly from its raw CSV file under `data/raw/` via a single shared `load_raw_dataset()` function, with no dataset-specific loading logic.

Table 1: Raw shape, feature-type composition, missingness, and target positive rate for the three datasets before any preprocessing.

Dataset	Rows	Numeric	Categorical	Missing	Positive rate
Dataset 1	445,132	6	34	38/40 columns	5.64%
Dataset 2	918	6	5	none	55.34%
Dataset 3	253,680	7	14	none	9.42%

Dataset 1 — Personal Key Indicators of Heart Disease (2022): sourced from the Kaggle 2022 BRFSS release [6] and distributed as `heart_2022_with_nans.csv`. This variant is used deliberately, rather than the pre-cleaned no-NaN release, so that the preprocessing pipeline demonstrates a realistic missing-value-handling workflow. The raw table has 445,132 rows and 40 columns: 34 categorical columns (`State`, `Sex`, `GeneralHealth`, and most of the Yes/No health-history flags such as `HadHeartAttack`, `HadDiabetes`, `HadStroke`) and 6 numeric columns (`PhysicalHealthDays`, `MentalHealthDays`, `SleepHours`, `HeightInMeters`, `WeightInKilograms`, `BMI`). The target column, `HadHeartAttack`, is a Yes/No self-report that is mapped to a binary 0/1 label during preprocessing; before cleaning, 93.67% of respondents (416,959) answered ”No”, 5.64% (25,108) answered ”Yes”, and 0.69% (3,065) left the field blank, giving a roughly 17:1 imbalance between the majority and minority classes once the missing-target rows are dropped.

Thirty-eight of the 40 columns contain at least one missing value; only `State` and `Sex` are fully populated. Table 2 lists the six most-affected columns, all vaccination/screening-history fields that respondents frequently skip or do not know the answer to. The core numeric health metrics have moderate missingness (`BMI` 10.96%, `WeightInKilograms` 9.45%, `HeightInMeters` 6.44%), while the disease-history Yes/No flags and core demographics are the most complete, at well under 1% missing — a pattern typical of self-reported survey data, where harder-to-recall or less commonly asked questions accumulate more gaps.

Table 2: The six most-missing columns in Dataset 1 (raw data, before imputation).

Column	Missing rate
TetanusLast10Tdap	18.54%
PneumoVaxEver	17.31%
HIVTesting	14.86%
ChestScan	12.59%
CovidPos	11.40%
HighRiskLastYear	11.37%

Dataset 2 — Heart Failure Prediction: sourced from the fedesoriano Kaggle release [7] and distributed as `heart.csv`. This is the smallest and already mostly-clean table used in this study: 918 rows and 12 columns, combining 5 clinical/demographic categorical columns (`Sex`, `ChestPainType`, `RestingECG`, `ExerciseAngina`, `ST_Slope`) with 6 numeric columns (`Age`, `RestingBP`, `Cholesterol`, `FastingBS`, `MaxHR`, `Oldpeak`) and the pre-encoded binary target `HeartDisease`. There are zero missing values across all 918 rows, so the dataset appears complete on the surface, but two of its numeric columns use `0` as an undocumented placeholder for "not measured" rather than a genuine reading: exactly 1 row has `RestingBP == 0` and 172 rows (18.7%) have `Cholesterol == 0`, far too many to be real zero-cholesterol readings given that a cholesterol or blood-pressure value of 0 is not physiologically possible. `DatasetConfig.zero_as_missing_columns` records both columns so that the preprocessing pipeline recodes these placeholder zeros as `NaN` before imputation rather than taking them at face value. Unlike Dataset 1, the target here is close to balanced: 508 patients (55.34%) have heart disease and 410 (44.66%) do not, which is typical of a clinical dataset curated specifically for a heart-disease classification task rather than drawn from a general population survey.

Dataset 3 — Heart Disease Health Indicators (BRFSS 2015): sourced from the alexteboul Kaggle release [8] and distributed as `heart_disease_health_indicators_BRFSS2015.csv`. The raw table has 253,680 rows and 22 columns, all already numerically encoded as `float64` – including conceptually binary flags (`HighBP`, `Smoker`, `Sex`) and small ordinal scales (`GenHlth` 1–5, `Age` 1–13 brackets, `Education` 1–6, `Income` 1–8). Seven columns (`BMI`, `GenHlth`, `MentHlth`, `PhysHlth`, `Age`, `Education`, `Income`) are treated as numeric and the remaining 14 as categorical. There are zero missing values anywhere in the dataset, making it the cleanest of the three in terms of completeness. However, 23,899 rows (9.42%) are exact duplicates of another row across all 22 columns – plausible here because every feature is binary or a small discrete scale, so two distinct respondents can easily share an identical combination of answers. This is a data-quality issue distinct from missingness, and `DatasetConfig` flags Dataset 3 with `drop_duplicate_rows=True` so these rows are removed before the target is cleaned (Section V.3.). The target, `HeartDiseaseorAttack`, is heavily imbalanced on the raw data: 229,787 respondents (90.58%) report no heart disease/attack history and 23,893 (9.42%) do, a roughly 1:9.6 minority-to-majority ratio, similar in shape to Dataset 1 but somewhat less extreme.

Across all three datasets, the target definition, raw imbalance, and data-quality quirks summarized above directly motivate the dataset-specific preprocessing decisions described in Section V.3.: the class-imbalance policy (F1), the zero-as-missing recoding restricted to Dataset 2, and the duplicate-row removal restricted to Dataset 3.

V.2. Exploratory Data Analysis

Before any preprocessing decision is made, each of the three raw datasets is examined independently in `notebooks/01_eda.ipynb` to characterize its class balance, missingness, numeric-feature relationships, and the strength of association between individual features and the target. The purpose of this stage is not to fit any model, but to surface the data-quality issues and predictive signals that later motivate specific preprocessing choices (see Section V.3.) and the class-imbalance policy adopted in the modeling stage (Section V.1. and finding F1 of the project's scope review).

Class imbalance. All three datasets exhibit some degree of class imbalance, but to very different extents. Dataset 1 is the most imbalanced, with a raw positive rate of 5.64%, giving balanced class weights of approximately $\{0 : 0.53, 1 : 8.80\}$ – a positive example is weighted roughly 16.6× more than a negative one under scikit-learn's balanced weighting scheme [9]. Dataset 3 is imbalanced in the same direction but less severely, with a raw positive rate of 9.42% and balanced weights of approximately $\{0 : 0.55, 1 : 5.31\}$ (about 9.6×). Dataset 2 is the outlier among the three: its positive rate of 55.34% is close to balanced, giving weights near 1.0 in both directions ($\{0 : 1.12, 1 : 0.90\}$) and requiring very little correction. This spread of imbalance regimes across the three datasets is precisely what motivates RQ1/RQ2's design: a class- imbalance policy that only worked well on a near-balanced dataset like Dataset 2 would not generalize to the 17:1 imbalance of Dataset 1.

Missingness. Dataset 1 is the only one of the three with meaningful missing data: 38 of its 40 columns contain at least one missing value, led by vaccination/screening-history fields that respondents often skip (Table 2). Datasets 2 and 3 report zero missing values on the surface, but Dataset 2's completeness is misleading – its `RestingBP` and `Cholesterol` columns use an undocumented `0` placeholder for "not measured" (1 and 172 rows respectively) that is not distinguishable from a genuine reading without domain knowledge. This is why the EDA stage treats "zero missing values" and "no data-quality issues" as two separate questions rather than assuming one implies the other.

Numeric feature correlations. Correlation matrices computed over each dataset's numeric columns show no evidence of problematic multicollinearity in any of the three datasets, though each has its own notable pairwise relationships. In Dataset 1, `WeightInKilograms` and `BMI` are strongly correlated ($r = 0.86$), which is expected since BMI is mathematically derived from weight and height; `HeightInMeters` and `WeightInKilograms` are moderately correlated ($r = 0.47$), while `HeightInMeters` and `BMI` are nearly uncorrelated ($r = -0.03$), confirming that BMI successfully normalizes out the effect of height. `PhysicalHealthDays` and `MentalHealthDays` are moderately correlated ($r = 0.32$). In Dataset 2, no pair of numeric features is strongly correlated; the strongest relationships are `Age` ↔ `MaxHR` ($r = -0.38$) and `Age` ↔ `Oldpeak` ($r = 0.26$), with `Cholesterol` weakly negatively correlated with `FastingBS` ($r = -0.26$), plausibly diluted by the placeholder-zero `Cholesterol` values noted above. In Dataset 3, `PhysHlth` and `GenHlth` are the most correlated pair ($r = 0.52$), `MentHlth` and `PhysHlth` are moderately correlated ($r = 0.35$), and `Education` and `Income` show the strongest socioeconomic correlation overall ($r = 0.45$). In all three cases, these pairwise relationships are informative but well below the 0.9 correlation threshold later used for feature reduction (Section V.3.), so they surface as candidates for the modeling stage rather than as columns to drop outright at the EDA stage.

Outlier detection (Dataset 2 only). Dataset 2 is the only dataset whose numeric columns are genuine continuous clinical measurements rather than mostly binary or discrete survey answers, which is why IQR (Tukey-fence) outlier detection is examined here and later applied only to Dataset 2 during preprocessing (finding F5). Computed on the full raw dataset for exploratory purposes, `Age` has no values outside its fence; `RestingBP` (bounds 90–170) flags 27 potential outliers; `Cholesterol` (bounds 105.6–376.6, computed after treating its zero-placeholders as missing) flags 23; `MaxHR` (bounds 66–210) flags 2; and `Oldpeak` (bounds −2.25–3.75) flags 16, consistent with its long right tail. These full-dataset bounds are for exploratory characterization only; the bounds actually used to clip Dataset 2's train/test splits in Section V.3. are refit on the training split alone to avoid leaking test-set extremes into the fences.

Target rate by category. Categorical features associated with self-reported general health status are, across every dataset that has one, by far the strongest single predictors observed during EDA. In Dataset 1, the heart-attack rate rises monotonically with self-reported `GeneralHealth` from 1.53% ("Excellent") to 2.84% ("Very good"), 5.78% ("Good"), 11.99% ("Fair"), and 21.08% ("Poor") – a roughly 14-fold spread across five categories, in contrast to BMI, whose distribution is nearly identical between positive and negative cases (median around 27–28 in both groups) and is therefore a comparatively weak discriminator on its own. In Dataset 2, both categorical predictors examined are strong: patients with asymptomatic chest pain (`ASY`) have a heart-disease rate of 79.03%, far higher than `ATA` (13.87%), `NAP` (35.47%), or `TA` (43.48%) – counterintuitively, the absence of typical chest-pain symptoms is the strongest single warning sign in this dataset – and a `Flat` (82.83%) or `Down` (77.78%) ST-segment slope is associated with a much higher rate than an `Up`-sloping segment (19.75%). In Dataset 3, the same `GenHlth` pattern recurs almost identically to Dataset 1 despite the two datasets being independent BRFSS survey years: the rate rises from 2.24% ("Excellent") to 4.63%, 10.46%, 21.31%, and 34.00% ("Poor"); `HighBP` is also strongly associated, with 16.47% of respondents with high blood pressure reporting heart disease/attack versus only 4.12% of those without.

Taken together, these EDA findings motivate three concrete preprocessing decisions detailed in Section V.3.: a shared class-imbalance policy applied at model-fit time rather than by resampling (F1, since the degree of imbalance varies too much across datasets for a single resampling ratio to be appropriate); IQR-based outlier clipping scoped to Dataset 2 only (F5, since Datasets 1 and 3 are mostly binary/discrete survey data where a genuinely high value is itself a disease signal rather than noise); and dataset-specific data-quality handling – zero-as-missing recoding for Dataset 2 and duplicate-row removal for Dataset 3 – rather than a single one-size-fits-all cleaning rule.

V.3. Data Preprocessing Pipeline

The preprocessing stage uses one shared pipeline with dataset-specific choices recorded in the `DatasetConfig` registry. The raw data are normalized, the target is cleaned and encoded as binary, and each dataset is split into a development partition and an untouched test partition before any learned transformation is fitted. The final experiment settings are read from `config.yaml`; dataset metadata remain in `src/config.py`.

Dataset-specific cleaning. Dataset 3 removes exact duplicate rows because its mostly binary and ordinal representation produces repeated response patterns. Dataset 2 converts zero values in `Cholesterol` and `RestingBP` to missing values because zero is not physiologically plausible for these measurements. Missing target values are removed before the binary target is encoded.

Learned transformations. Numeric values are median-imputed and standardized; categorical values are mode-imputed, binary-encoded, or one-hot encoded. Dataset 2 additionally uses training-only IQR clipping on its continuous clinical measurements. A variance filter and correlation filter are then fit on processed training features. These transformations are fitted on fold-training data inside cross-validation and on the full development split for the final hold-out evaluation; the test rows never determine fitted statistics.

Class imbalance is handled at model-fitting time with balanced sample weights, not by resampling the data during preprocessing. This preserves the natural class prevalence for validation and test metrics. The final processed shapes and row counts are summarized in Table 3.

Table 3: Final processed train/test row counts and feature counts, after feature reduction, for the three datasets.

Dataset	Train rows	Test rows	Final features (excl. target)
Dataset 1	353,653	88,414	131
Dataset 2	734	184	18
Dataset 3	183,824	45,957	22

The processed train/test files and the fold manifest are persisted under `data/processed/<dataset>/`. The same fold manifest is reused by the model and SHAP runners, while the fold-local fitting procedure is described in the Model Experimental Design and SHAP Methodology subsections.

V.4. Model Experimental Design

The modeling stage consumes the processed development and test files produced by the upstream data pipeline. For each of the three datasets, the data pipeline creates a stratified 80/20 development–test split. The test partition is held out during model comparison and is used only after cross-validation has been completed. All three candidate algorithms use the same raw development rows and persisted fold assignments for a given dataset. Within each fold, preprocessing is fit only on the fold-training rows and then applied to the validation rows, so model differences are not confounded by different input representations or validation-fold preprocessing statistics.

Within the development partition, we apply stratified five-fold cross-validation with shuffling and a fixed random seed of 42. Stratification preserves approximately the same class proportion in each fold, which is particularly important when the positive healthcare outcome is less frequent [9]. The exact same fold-generation rule and random seed are used for AdaBoost, XGBoost, and LightGBM. For every fold, four folds are used for model fitting and the remaining fold is used for validation. Model objects and preprocessing objects are re-created from scratch for every fold to prevent fitted state or validation statistics from leaking between folds.

To reduce the effect of class imbalance during fitting, balanced sample weights are computed from the training portion of each fold only. Validation and test sets remain untouched so that the reported metrics reflect the natural class prevalence of each dataset. A fixed probability threshold of 0.50 is used for the threshold-dependent metrics. No threshold is optimized on the validation folds, which avoids adding a second tuning procedure that could make fold comparisons less consistent.

V.5. Model Configuration and Hyperparameters

The core comparison uses one fixed, literature-guided configuration per algorithm across all three datasets. We intentionally do not select a different best hyperparameter configuration for each dataset using the same five folds that are later used to quantify stability. Doing so could make the reported cross-validation scores optimistic and would confound RQ2 with hyperparameter search variability. The selected settings use conservative learning rates, moderate ensemble sizes, and regularization/subsampling for the gradient- boosted models.

Table 4: Fixed model configurations used for the cross-dataset comparison.

Model	Selected hyperparameters	Rationale
AdaBoost	Decision stump (max depth = 1); 200 estimators; learning rate = 0.10	Decision stumps are the standard weak learner for AdaBoost. The larger ensemble combined with a smaller learning rate provides gradual boosting while limiting the contribution of each individual stump.
XGBoost	300 estimators; learning rate = 0.05; max depth = 4; subsample = 0.80; column sample = 0.80; L_2 regularization = 1; histogram tree method	A low learning rate and moderate tree depth control model complexity. Row and column subsampling reduce variance, while L_2 regularization makes leaf weights more conservative. The histogram method improves efficiency on the larger datasets.
LightGBM	300 estimators; learning rate = 0.05; 31 leaves; max depth = 6; minimum child samples = 20; subsample = 0.80; column sample = 0.80; L_2 regularization = 1	The number of leaves and depth cap constrain the leaf-wise tree growth. The minimum child size, subsampling, and regularization reduce overfitting while the small learning rate supports a smoother boosting process. Deterministic CPU settings and a fixed seed are enabled for reproducibility.

V.6. Evaluation Metrics and Model Comparison

Each validation fold is evaluated using ROC-AUC, PR-AUC, Recall, and F1-score. ROC-AUC measures ranking performance across classification thresholds. PR-AUC focuses on the precision–recall trade-off and is especially informative when the positive class is relatively rare [10]. Recall measures the proportion of positive cases correctly identified, while F1-score balances precision and recall at the fixed 0.50 threshold. Because the three datasets may have different class prevalence, PR-AUC is used as the primary ranking metric, with ROC-AUC, F1-score, and fold variability retained for a more complete comparison.

For RQ1, we report the mean five-fold score of every metric for every model–dataset pair and rank the three models within each dataset by mean PR-AUC. We additionally report the average within-dataset rank across all three datasets rather than pooling patient rows across datasets. This prevents the largest dataset from dominating the overall comparison.

For RQ2, predictive stability is measured directly from the five fold-level scores. For each metric we report the mean, standard deviation, minimum, maximum, and range across folds. A smaller standard deviation and range indicate that the model is less sensitive to the particular development-data partition. After cross-validation, each model is re-fitted once on the full development partition and evaluated on the untouched test partition. These test results are reported separately from the cross-validation results and are not used to select the best model.

The complete experiment configuration is stored in `config.yaml`. The implementation writes fold-level metrics, aggregated cross-validation summaries, held-out test metrics, per-dataset rankings, cross-dataset rankings, and an experiment metadata file containing package versions and a hash of the configuration. This makes the model experiments repeatable and provides the fold-level outputs required by the later SHAP stability analysis. “‘latex

V.7. SHAP Explainability and Feature-Ranking Stability (RQ3)

To address RQ3, this study uses SHAP (SHapley Additive exPlanations) to examine both the importance of individual input features and the stability of these explanations across cross-validation folds. SHAP assigns an additive contribution value to each feature for an individual prediction and is grounded in Shapley-value principles from cooperative game theory [4]. For tree-based models, TreeSHAP provides an efficient method for estimating these feature attributions [5].

The SHAP analysis follows the same predefined stratified five-fold partitions used in the predictive experiments. These fold assignments are generated by the data-preprocessing pipeline and stored in `data/processed/<dataset>/kfold_indices.csv`. Reusing the same folds ensures that predictive stability and explanation stability are evaluated under identical train–validation partitions.

For each dataset, model, and fold, the model is trained on four folds and SHAP values are computed on a deterministic sample from the held-out validation fold. This procedure prevents the explained observations from being reused as training samples and produces one independent feature-importance ranking for each fold. The random seed, maximum number of explained validation samples, background size, and top-k value are controlled through `config.yaml` to ensure reproducibility.

For XGBoost and LightGBM, SHAP values are computed using `TreeExplainer`. The scikit-learn implementation of AdaBoost used in this study is not supported by TreeExplainer in the pinned SHAP version. Therefore, AdaBoost is explained using model-agnostic permutation SHAP with a deterministic background sample drawn from the corresponding fold-training data. Because the underlying explainer differs between AdaBoost and the gradient-boosted models, raw SHAP magnitudes are not compared directly across model families. Instead, the stability analysis focuses on how each model's own feature ranking changes across folds.

For a feature j in fold f, global SHAP importance is defined as the mean absolute SHAP value across the N_f explained validation observations:

$$I_{j,f} = \frac{1}{N_f} \sum_{i=1}^{N_f} |\phi_{i,j,f}|, \tag{1}$$

where $\phi_{i,j,f}$ denotes the SHAP attribution of feature j for observation i in fold f. Mean absolute SHAP is used because positive and negative contributions should not cancel when measuring overall feature importance. Features are sorted in descending order of $I_{j,f}$ to construct a fold-specific feature ranking.

To evaluate the robustness of these rankings, three complementary stability metrics are used. First, Kendall's rank correlation coefficient τ measures the agreement between two feature rankings based on concordant and discordant feature pairs. Values closer to one indicate stronger agreement in pairwise feature ordering.

Second, Spearman's rank correlation coefficient ρ measures monotonic agreement between the complete rank vectors of two folds. A value close to one indicates that the overall ordering of features remains highly consistent between folds.

Third, stability of the most influential features is measured using top-k Jaccard similarity:

$$J(A, B) = \frac{|A \cap B|}{|A \cup B|}, \tag{2}$$

where A and B represent the sets of the top-k ranked features from two cross-validation folds. The experiment uses $k = 10$. A Jaccard score of one indicates identical top-10 feature membership, whereas a lower score indicates that some features enter or leave the top-10 across folds.

With five cross-validation folds, ten unique fold pairs are evaluated for every dataset–model combination. Kendall's τ and Spearman's ρ assess stability of the complete feature ranking, while top-10 Jaccard similarity focuses specifically on the features considered most important.

A consensus ranking is additionally constructed by aggregating fold-level results. For each feature, the analysis records its mean absolute SHAP value, average rank, rank standard deviation, and frequency of appearing among the top-10 features. This provides a summary of both average feature importance and the degree to which individual features maintain their importance across different data partitions.

Several visual analyses are generated to complement the numerical stability metrics. SHAP beeswarm plots visualize the distribution and direction of feature effects across held-out validation observations, while mean absolute SHAP bar plots summarize global feature importance. Violin summary plots provide an additional view of the distribution of SHAP contributions for the most important features.

For stability analysis, feature-ranking heatmaps display the position of the leading features across the five folds. These plots make rank changes directly visible and help identify features whose importance is sensitive to the training partition. Fold-correlation heatmaps visualize the pairwise Spearman correlation between fold-specific feature rankings, making it possible to identify individual folds that differ from the remaining partitions.

The implementation stores fold-level SHAP importance values, pairwise stability measurements, aggregated stability statistics, consensus rankings, and the generated explainability and stability plots. The corresponding empirical results are presented in Section VI.3.. “‘

VI. Results and Discussion

VI.1. Predictive Performance Across Datasets (RQ1)

Table 5 summarizes the five-fold cross-validation performance of AdaBoost, XGBoost, and LightGBM on the three healthcare datasets. The values are reported as mean ± standard deviation across the five validation folds. PR-AUC is treated as the primary comparison metric, while ROC-AUC, Recall, and F1-score provide complementary views of ranking and threshold-based performance.

Table 5: Five-fold cross-validation performance (mean ± standard deviation).

extbfDataset	**Model**	**ROC-AUC**	**PR-AUC**	**Recall**	**F1**
Dataset 1	AdaBoost	0.8715 ± 0.0042	0.3853 ± 0.0087	0.6794 ± 0.0106	0.3765 ± 0.0082
	XGBoost	0.8873 ± 0.0031	0.4138 ± 0.0092	0.7799 ± 0.0071	0.3328 ± 0.0023
	LightGBM	0.8871 ± 0.0034	0.4127 ± 0.0090	0.7760 ± 0.0097	0.3347 ± 0.0029
Dataset 2	AdaBoost	0.9240 ± 0.0429	0.9228 ± 0.0438	0.8449 ± 0.0294	0.8587 ± 0.0169
	XGBoost	0.9274 ± 0.0332	0.9248 ± 0.0338	0.8794 ± 0.0451	0.8780 ± 0.0234
	LightGBM	0.9249 ± 0.0303	0.9169 ± 0.0329	0.8818 ± 0.0235	0.8787 ± 0.0169
Dataset 3	AdaBoost	0.8324 ± 0.0031	0.3678 ± 0.0078	0.7727 ± 0.0082	0.3826 ± 0.0030
	XGBoost	0.8386 ± 0.0026	0.3803 ± 0.0050	0.8014 ± 0.0075	0.3805 ± 0.0022
	LightGBM	0.8378 ± 0.0023	0.3797 ± 0.0059	0.7955 ± 0.0047	0.3822 ± 0.0021

On Dataset 1, XGBoost and LightGBM produced almost identical ROC-AUC, while XGBoost achieved a slightly higher mean PR-AUC and Recall. Both gradient- boosted models clearly exceeded AdaBoost on ROC-AUC, PR-AUC, and Recall. The lower F1 scores of XGBoost and LightGBM indicate that their higher recall at the fixed 0.50 threshold came with a precision trade-off.

On Dataset 2, all three models achieved strong discrimination, with mean ROC-AUC values close to 0.924. XGBoost achieved the highest mean Recall and F1, whereas AdaBoost obtained the highest mean PR-AUC by a small margin. This shows that no single model dominated every metric on this relatively small dataset.

On Dataset 3, XGBoost and LightGBM again improved ROC-AUC and PR-AUC over AdaBoost. XGBoost achieved the highest mean Recall, while the three models had very similar F1 values. Across the three datasets, the aggregated comparison ranked XGBoost first with a mean dataset rank of 1.0000, followed by LightGBM at 2.3333 and AdaBoost at 2.6667. XGBoost also achieved the highest cross-dataset mean ROC-AUC (0.8844), PR-AUC (0.5730), and Recall (0.8202). AdaBoost achieved the highest cross-dataset mean F1 (0.5392), emphasizing that the preferred model depends partly on the evaluation objective and decision threshold.

Table 6: Unweighted cross-dataset comparison based on the experiment outputs.

Model	Mean Rank	ROC-AUC	PR-AUC	Recall	F1
XGBoost	**1.0000**	**0.8844**	**0.5730**	**0.8202**	0.5305
LightGBM	2.3333	0.8833	0.5698	0.8178	0.5319
AdaBoost	2.6667	0.8760	0.5586	0.7657	**0.5392**

These results answer RQ1 by showing that XGBoost provides the strongest overall performance under the selected protocol, although LightGBM is extremely close on several ranking metrics and AdaBoost retains a small advantage in aggregate F1. Therefore, the comparison does not support a claim that one algorithm is uniformly superior on every metric and dataset.

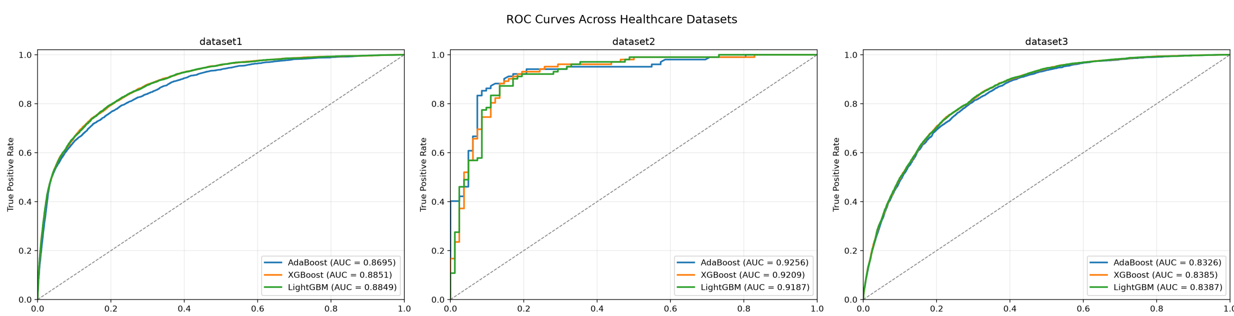

Figure 3: ROC curves for the three models on the untouched hold-out test sets across the three healthcare datasets.

The ROC visualization is consistent with the tabulated test results. The curves are close on Dataset 1 and Dataset 3, where XGBoost and LightGBM provide similar ranking performance. Dataset 2 has the highest test AUC values overall, but its step-shaped curves also reflect the smaller test sample and therefore should not be interpreted as evidence of uniformly better clinical performance across datasets. The figure supports comparing ranking behavior within each dataset, not pooling patients from different prediction tasks.

Table 7: Performance on the untouched hold-out test partitions after refitting on the full development data.

$\rho = 0.9324$ but Kendall $\tau = 0.7990$. Both values indicate positive and

Dataset	Model	ROC-AUC	PR-AUC	Recall	F1
Dataset 1	AdaBoost	0.8695	0.3860	0.6705	0.3657
	XGBoost	0.8850	0.4169	0.7712	0.3255
	LightGBM	0.8850	0.4187	0.7694	0.3259
Dataset 2	AdaBoost	0.9256	0.9394	0.8529	0.8832
	XGBoost	0.9224	0.9216	0.8333	0.8586
	LightGBM	0.9189	0.9166	0.8431	0.8643
Dataset 3	AdaBoost	0.8326	0.3672	0.7674	0.3788
	XGBoost	0.8384	0.3756	0.7963	0.3773
	LightGBM	0.8385	0.3763	0.7925	0.3789

VI.2. Predictive Stability Across Cross-Validation Folds (RQ2)

The fold-level results also reveal clear differences in stability across the three datasets. Dataset 1 and Dataset 3 produced small standard deviations for all three models. For example, XGBoost obtained ROC-AUC standard deviations of 0.0033 on Dataset 1 and 0.0027 on Dataset 3. LightGBM showed similarly small variation, indicating that the two gradient-boosted models were not highly sensitive to the particular fold assignment on the two larger datasets.

Dataset 2 exhibited noticeably greater fold-to-fold variation. Its development partition contains only 734 samples, compared with 353,653 for Dataset 1 and 183,824 for Dataset 3. XGBoost ROC-AUC ranged from 0.8676 to 0.9465 across the five folds, while AdaBoost ranged from 0.8547 to 0.9673 and LightGBM ranged from 0.8651 to 0.9497. The larger variability is therefore consistent with the smaller sample size: each validation fold contains fewer observations, so the composition of a single fold has a greater influence on the measured score.

The final hold-out results were generally close to the corresponding cross-validation means. For example, Dataset 1 XGBoost achieved a mean CV ROC-AUC of 0.8873 and a hold-out ROC-AUC of 0.8851, while Dataset 3 XGBoost achieved 0.8386 in CV and 0.8385 on the hold-out set. This agreement provides additional evidence that the reported cross-validation estimates are not driven by a single unusually favorable split.

Overall, RQ2 is answered by finding that predictive performance is highly stable on the two large datasets but less stable on the small Heart Failure Prediction dataset. The pattern is observed across all three algorithms, suggesting that dataset size and fold composition contribute more to the observed instability than the choice among the three boosting methods alone.

VI.3. SHAP Feature-Importance Stability Across Cross-Validation Folds (RQ3)

Following the fold-wise explainability procedure described in Section V.7., RQ3 evaluates whether SHAP-based global feature-importance rankings remain consistent when the training and validation partitions change across cross-validation folds. For every dataset–model combination, all ten unique pairs of the five folds are compared using Kendall's τ, Spearman's ρ, and top-10 Jaccard similarity.

Table 8 shows that the SHAP feature rankings are generally stable across folds for all nine dataset–model combinations. Mean Spearman correlations range from 0.9324 to 0.9921, indicating strong agreement in the complete feature ordering. Mean Kendall coefficients range from 0.7990 to 0.9636, providing additional evidence that most pairwise feature ordering relationships are preserved across folds. Top-10 Jaccard similarity ranges from 0.7758 to 1.0000, indicating that the most influential feature sets are also largely consistent.

Table 8: Mean SHAP feature-ranking stability across the ten pairwise comparisons of five cross-validation folds.

Dataset	**Model**	**Kendall τ**	**Spearman ρ**	**Top-10 Jaccard**
Dataset 1	AdaBoost	0.9593	0.9644	0.9273
	XGBoost	0.7990	0.9324	0.8273
	LightGBM	0.8057	0.9368	0.7758
Dataset 2	AdaBoost	0.8536	0.9410	0.8061
	XGBoost	0.8118	0.9346	0.8545
	LightGBM	0.8301	0.9422	0.8364
Dataset 3	AdaBoost	0.9636	0.9921	**1.0000**
	XGBoost	0.9255	0.9814	**1.0000**
	LightGBM	0.8996	0.9735	0.9273

Dataset 3 demonstrates particularly strong explanation stability. AdaBoost and XGBoost both obtain a mean top-10 Jaccard score of 1.0000, indicating that the same set of ten most influential features is retained across every fold pair. LightGBM achieves the highest overall mean Spearman correlation of 0.9735, showing that its complete feature ranking is also highly consistent. The high Kendall coefficients for all three models further confirm that the relative ordering of feature pairs changes only modestly across partitions.

Dataset 1 also exhibits strong overall stability, although the different metrics reveal some variation near the top of the ranking. AdaBoost achieves the strongest agreement on this dataset, with mean Kendall $\tau = 0.9593$, Spearman $\rho = 0.9644$, and top-10 Jaccard similarity of 0.9273. LightGBM, however, obtains a high Spearman correlation of 0.9368 while its top-10 Jaccard similarity is lower at 0.7758. This result indicates that the complete ranking remains similar across folds even though several features near the top-10 boundary exchange positions or move into and out of the top feature set.

The difference between the Kendall and Spearman measurements is also informative. For example, Dataset 1 XGBoost achieves Spearman $\rho = 0.9324$ but Kendall $\tau = 0.7990$. Both values indicate positive and strong ranking agreement, but Kendall's lower value suggests that some pairwise feature orderings change across folds even though the overall ranking structure remains similar. Reporting both statistics therefore provides a more complete view of explanation robustness than relying on a single correlation measure.

Dataset 2 remains reasonably stable despite being substantially smaller than Datasets 1 and 3. All three models obtain mean Spearman correlations greater than 0.93, while mean Kendall coefficients range from 0.8118 to 0.8536. Top-10 Jaccard similarity ranges from 0.8061 to 0.8545. These values indicate that the majority of important features remain consistently ranked even though some movement occurs between folds.

The SHAP beeswarm plots complement the global importance rankings by showing both the magnitude and direction of feature contributions. Features appearing near the top of the beeswarm plots have large absolute effects on the model output, while the horizontal distribution of SHAP values indicates whether particular feature values tend to increase or decrease the predicted risk. The corresponding mean absolute SHAP bar plots provide a simpler view of global importance, and the violin plots show the distribution of SHAP effects for the leading features.

The feature-ranking heatmaps provide a direct visualization of explanation stability across folds. Features whose heatmap cells remain at similar rank values across all five folds can be considered highly stable, whereas large changes in rank indicate sensitivity to the particular training partition. This visualization is especially useful for distinguishing small changes near the top-10 cutoff that may reduce Jaccard similarity without substantially changing the complete ranking.

The fold-correlation heatmaps provide an additional model-level view of stability. High off-diagonal Spearman correlations indicate that different training folds produce similar global feature orderings. A fold with substantially lower correlations to the other folds would suggest that the corresponding training partition produces a noticeably different explanation. Together with the ranking heatmaps, these plots allow the numerical stability scores to be visually inspected rather than interpreted only from aggregated statistics.

Overall, RQ3 is answered by finding that SHAP-based global feature-importance rankings are robust to cross-validation partitioning for the evaluated boosting models. This conclusion is supported by three complementary measures: Kendall's τ shows strong pairwise ordering agreement, Spearman's ρ shows high consistency in complete feature rankings, and top-10 Jaccard similarity shows that the most influential feature sets are generally preserved.

At the same time, the results demonstrate why multiple stability metrics are necessary. A model can maintain a high full-ranking correlation while still showing changes in the membership of its top-10 features. Consequently, explanation stability should not be judged from a single correlation coefficient alone. Combining rank correlations, top-feature overlap, feature-ranking heatmaps, and fold-correlation heatmaps provides a more complete assessment of the reproducibility of model explanations.

The observed SHAP stability should be interpreted as evidence of explanation robustness rather than causal validity. A feature that is consistently important across folds is repeatedly used by the model under the observed data distribution, but this does not imply that changing that feature would causally alter a patient's health outcome. In addition, because AdaBoost uses permutation SHAP while XGBoost and LightGBM use TreeSHAP, raw SHAP magnitudes are not used for direct cross-model comparisons. RQ3 therefore focuses on within-model fold-to-fold ranking stability, which remains comparable across the experimental protocol.

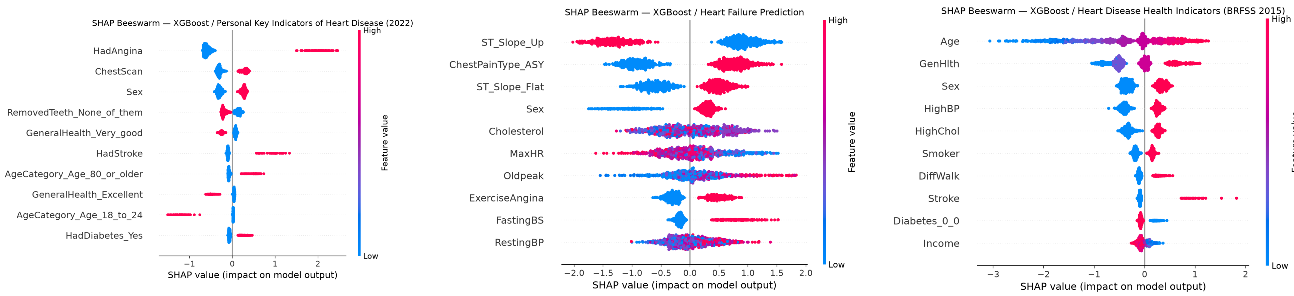

Figure 4: SHAP beeswarm plots for XGBoost on Dataset 1, Dataset 2, and Dataset 3, respectively.

The beeswarm plots show that the most influential features are dataset-specific. For Dataset 1, `HadAngina`, `ChestScan`, and `Sex` occupy the leading positions. Dataset 2 is dominated by `ST_Slope_Up`, `ChestPainType_ASY`, and `ST_Slope_Flat`, whereas Dataset 3 is led by `Age`, `GenHlth`, and `Sex`. The horizontal position indicates contribution direction and magnitude, while color encodes the feature value. These patterns show why a single cross-dataset feature interpretation would be misleading even when the same model family is used.

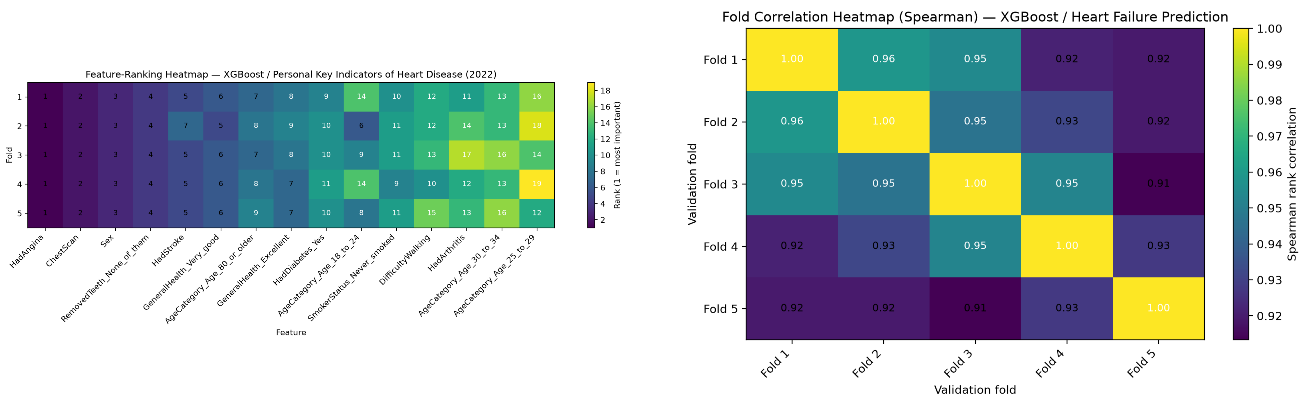

Figure 5: Representative SHAP stability visualizations: feature-ranking heatmap for Dataset 1 and fold-correlation heatmap for Dataset 2 using XGBoost.

The ranking heatmap makes the fold-level movement of individual features visible, while the correlation heatmap summarizes agreement between complete rankings. The high off-diagonal correlations in the Dataset 2 example are consistent with its mean Spearman value of 0.9346, even though its smaller sample produces larger predictive variability. Together, the plots clarify that explanation stability concerns reproducibility of model rankings, not causal importance or clinical validity.

VI.4. Discussion

Taken together, the three research questions provide complementary evidence about predictive quality and interpretability. RQ1 shows that XGBoost provides the strongest overall predictive ranking across the three datasets, with LightGBM very close on ROC-AUC and PR-AUC and AdaBoost retaining the highest cross-dataset mean F1. RQ2 shows that predictive scores are highly stable on the two large datasets but vary more on the smaller Heart Failure Prediction dataset. RQ3 extends the stability analysis from prediction metrics to model explanations and finds that the SHAP feature rankings remain highly consistent across folds for all three boosting methods.

The SHAP results should be interpreted as measures of explanation stability, not as evidence of causality. A feature can be consistently important to a model because it is predictive under the observed data distribution, but this does not establish that changing the feature would causally change a patient's risk. In addition, raw SHAP magnitudes are not compared directly between AdaBoost and the two gradient-boosted models because AdaBoost is explained with permutation SHAP whereas XGBoost and LightGBM use TreeSHAP. The study therefore bases RQ3 on within-model fold-to-fold ranking consistency, which is comparable across the experimental protocol.

Overall, the combined findings indicate that the strongest models are not only competitive in predictive performance but also produce feature-importance patterns that are reproducible across data partitions. This is useful for healthcare risk-prediction studies because an explanation that changes sharply with a different fold would be difficult to trust even if average predictive performance were high. The observed stability therefore provides an additional criterion, alongside ROC-AUC, PR-AUC, Recall, and F1, for assessing the robustness of the evaluated models.

VII. Conclusion and Future Work

VII.1. Conclusion

This study compared AdaBoost, XGBoost, and LightGBM for binary healthcare risk prediction across three datasets using a common experimental protocol. The protocol combined stratified development–test separation, fixed five-fold partitions, fold-local preprocessing, balanced training weights, multiple evaluation metrics, and fold-wise SHAP analysis. This design allowed predictive performance and explanation stability to be examined under the same data partitions rather than inferred from unrelated experimental settings.

For RQ1, XGBoost achieved the strongest overall ranking across the three datasets, with mean dataset rank 1.0000, mean ROC-AUC 0.8844, mean PR-AUC 0.5730, and mean Recall 0.8202. However, AdaBoost achieved the highest cross-dataset mean F1-score (0.5392), and the hold-out results show that the best model can depend on the dataset and metric. Therefore, the results support XGBoost as the strongest overall model under the selected ranking-oriented protocol, not as a universally superior healthcare model.

For RQ2, predictive performance was relatively stable on the two large datasets but more variable on the smaller Heart Failure Prediction dataset. This pattern was observed across the models and is consistent with the greater influence of fold composition when fewer observations are available. For RQ3, SHAP rankings were generally stable: mean Spearman correlations ranged from 0.9324 to 0.9921, and mean top-10 Jaccard similarity ranged from 0.7758 to 1.0000. The explanation results therefore support the robustness of the observed feature rankings across the evaluated partitions, while also showing that full-ranking correlation and top-feature overlap capture different aspects of stability.

The conclusions have several limitations. The study uses three public datasets and fixed, literature-guided hyperparameters rather than dataset-specific tuning. The datasets represent different prediction tasks and populations, so cross-dataset averages should be interpreted as unweighted summaries rather than evidence of clinical generalization. In addition, SHAP importance describes model behavior under the observed data distribution; it does not establish causal effects or clinical usefulness. Finally, the study evaluates stability across one five-fold partition and does not replace external validation on a new population.

VII.2. Future Work

Future work should first test whether the stability patterns persist under repeated stratified cross-validation with multiple random partitions. This would distinguish conclusions that are robust to the fold assignment from conclusions specific to the current five-fold split. A second direction is external validation on healthcare datasets from different institutions or collection periods, together with calibration and decision-curve analysis so that ranking performance is complemented by clinically relevant probability assessment.

Further work should also analyze performance and explanation stability across demographic or clinical subgroups. This would help determine whether a high aggregate score hides weaker performance or less consistent explanations for particular populations. Finally, model-specific tuning can be studied with nested cross-validation when the research objective changes from controlled comparison to model selection. Such tuning should be evaluated separately from the current fixed-configuration experiment so that hyperparameter-selection variability is not confused with fold stability.

VIII. References

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Appendix

1. **AIVIETNAM:** Further information about AIVIETNAM can be found here.
2. **GitHub Repository:**
3. **Dataset:**
4. **Demo Video / Presentation:**
    - Link demo video: Link demo
    - Link slide presentation: Link slide presentation